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An expert system based on Wavelet Neural Network-Adaptive Norm Entropy for scale invariant texture classification

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Abstract

Nowadays, texture classification becomes more important, as the computational power increases. The most important hardness of texture image analysis in the past was the deficiency of enough tools to characterize variety scales of texture images effectively. Recently, multi-resolution analysis such as Gabor filters, wavelet decompositions provide very good multi-resolution analytical tools for different scales of texture analysis and classification. In this paper, a Wavelet Neural Network based on Adaptive Norm Entropy (WNN-ANE) expert system is used for increasing the effectiveness of the scale invariant feature extraction algorithm (Best Wavelet Statistical Features (WSF)–Wavelet Co-occurrence Features (WCF)). Efficiently of proposed method was proved using exhaustive experiments conducted with Brodatz texture images.
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Keywords: Expert systems; Texture image; Wavelet statistical features; Wavelet co-occurrence features; Feature extraction; Texture classification

1. Introduction

Extracting invariant texture image features is an important issue in content-based image analysis (Pun & Lee, 2004). Texture is a low-level image feature. It is around all of us. There are many different applications involving texture analysis, including medical imaging, industrial inspection, document segmentation, radar image recognition, and texture-based image retrieval, etc. (Tuceryan & Jain, 1993). The statistical approaches to texture image analysis use the statistical definitions to characterize the texture as smooth, coarse, grainy, etc. (Conners & Harlow, 1980). The stochastic models such as Gaussian Markov random fields (GRMFs) and autoregression are used in texture analysis (Bovik, Clark, & Geisler, 1990). The last developments in the spatial/frequency analysis such as Gabor filters (Chang & Kuo, 1993; Teuner, Pichler, & Hosticka, 1995), wavelet decompositions (Laine & Fan, 1993; Pun & Lee, 2003; Unser, 1995) provide very good multi-resolution analytical tools for texture analysis and classification (Chang & Kuo, 1993). The application studies deal with these approaches show that they can achieve a high accuracy rate (Chang & Kuo, 1993).

So far many different approximations have been proposed, but most of these approximations assumed that the texture images have the same orientation and scale.

Nevertheless, this assumption is not valid for most practical applications (Chang & Kuo, 1993). Because, images may have different scales. The performances of above approximations become worse when the underlying assumption is no longer valid (Chang & Kuo, 1993).

For texture classification, recognition, and segmentation, proper attributes are required (Arivazhagan & Ganesan, 2003). So far, different feature extraction and classification approximations have been suggested for texture classification and segmentation. The oldest feature extraction methods were based on the first and second orders statistics of texture images (Chen & Pavlidis, 1983; Davis, Johns, & Aggarwal, 1979; Faugeras & Pratt, 1980; Haralick, Shanmugam, & Dinstein, 1973; Weszka, Dyer, & Rosenfeld, 1976). Gaussian Markov Random Fields

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(GMRF) and Gibbs Random Fields (GRF) were used for classification textures (Chellappa & Chatterjee, 1986; Cohen, Fan, & Patel, 1991; Cross & Jain, 1983; Derin & Elliot, 1987; Kashyap & Khotanzed, 1986; Manjunath & Chellappa, 1991). Local linear transformations are proposed to obtain texture image features (Laws, 1980; Unser, 1986). Haralick has firstly proposed co-occurence matrix features (Haralick et al., 1973). Co-occurence matrix has 14 features are to be computed that too for different distances at variety orientations which increases the computational and time complexity (Arivazhagan & Ganesan, 2003). Even all of co-occurence matrix features are used, the correct classification rate of 60–70% was only reported in the texture image literature.

The join drawback of these traditional statistical approximations such as one and second orders statistics, GMRF, GRF, co-occurence matrix and local linear transforms to texture analysis is being restricted to the analysis of spatial interactions on a single scale.

Nowadays, most popular methods on texture analysis are multi-resolution or multichannel analysis such as wavelet decompositions and Gabor filters have used (Bovik et al., 1990; Chang & Kuo, 1993; Haley & Manjunath, 1995; Manjunath & Ma, 1996; Raghu & Yegnanarayana, 1996; Unser, 1995; Unser & Eden, 1989; Vande Wouwer, Schenders, & Van Dyek, 1999; Wu & Wei, 1996). The wavelet transforms has more advantages than Gabor filters. Disadvantage of the Gabor filters is that the output of Gabor filter banks are not mutually orthogonal, which may cause an important correlation between texture image features (Arivazhagan & Ganesan, 2003). Wavelet and Gabor transforms are usually not reversible that restricts their applicability for texture retrieval, but wavelet transform can overcome some of these disadvantages. Wavelet transform is more superior than Gabor transform. Because wavelet transform provides a true and join frame work for the processing of a signal and image at variety scales (Arivazhagan & Ganesan, 2003; Unser, 1995). Moreover, the low pass and high pass filters used in the wavelet transform remain the same between two sequence scales while Gabor transform requires filters different parameters (Arivazhagan & Ganesan, 2003; Chang & Kuo, 1993). This status is another one disadvantage of Gabor filters. Because Gabor filter parameters need proper tuning of at variety scales (Arivazhagan & Ganesan, 2003).

The aims of this application can summarize as follow:

1. The effectiveness of the Discrete Wavelet Transform (DWT) features is shown to be used for texture image classification.
2. Obtaining of the co-occurrence matrix features is explained for texture image classification and sub-bands of wavelet transformed images.
3. A Wavelet Neural Network based on Adaptive Norm Entropy (WNN-ANE) algorithm is used for increasing the effectiveness of the scale invariant feature extraction algorithm (Best Wavelet Statistical Features (WSF)–Wavelet Co-occurrence Features (WCF)) presented in Arivazhagan and Ganesan (2003).
4. The correct texture classification performances of variety feature vectors obtained by using different wavelet families are compared.

In this study, experiments are conducted with 25 monochrome texture images, each of size 512×512, obtained from Brodatz image database in Fig. 1. The Discrete Wavelet Transform (DWT) is applied on a set of Brodatz texture image and statistical features such as mean, standard deviation, and norm entropy are extracted from the approximation and detail coefficients of DWT decomposed images, at variety scales. The different combinations of WNN-ANE improved in this study and WSF–WCF feature extraction method presented in Arivazhagan and Ganesan (2003) algorithms are applied for texture image classification by using variety wavelet families. The best feature vectors are chosen. Adaptive Norm Entropy values of the approximation and detail sub-bands of i-level ($i = 1, 2, 3$ respectively) DWT decomposed texture images are counted. The co-occurrence matrix of the approximation and detail sub-bands of 1-level DWT decomposed texture images are calculated for improve the success rate of texture classification. All of these features are given to inputs of a MultiLayer Perceptron Neural Network based on adaptive at classification stage. Other words, texture classification method improved in this paper consist of two stage: (1) feature extraction, (2) classification.

It is found that WNN-ANE improved in this study is superior than scale invariant feature extraction algorithm presented in Arivazhagan and Ganesan (2003) in point of the success rate of the Brodatz texture classification.

This paper is organized as follows: In Section 2, the theory of the pattern recognition, DWT, and Wavelet Neural Network are briefly reviewed. In Section 3, the feature extraction and texture classification are explained. In Section 4, the texture classification experimental results using different feature sets by using variety wavelet families are discussed in detail. In Section 5, concluding remarks are given.

2. Theoric informations

2.1. Pattern recognition

Pattern recognition can be divided into a sequence of stages, starting with feature extraction from the occurring patterns, which is the conversion of patterns to features that are regarded as a condensed representation, ideally containing all the necessary information. In the next stage, the feature selection step, a smaller number of meaningful features that best represent the given pattern without redundancy are identified. Finally, classification is carried out: a specific pattern is assigned to a specific class according to its characteristic features, selected for it. This general abstract model, which is demonstrated in Fig. 2, allows a

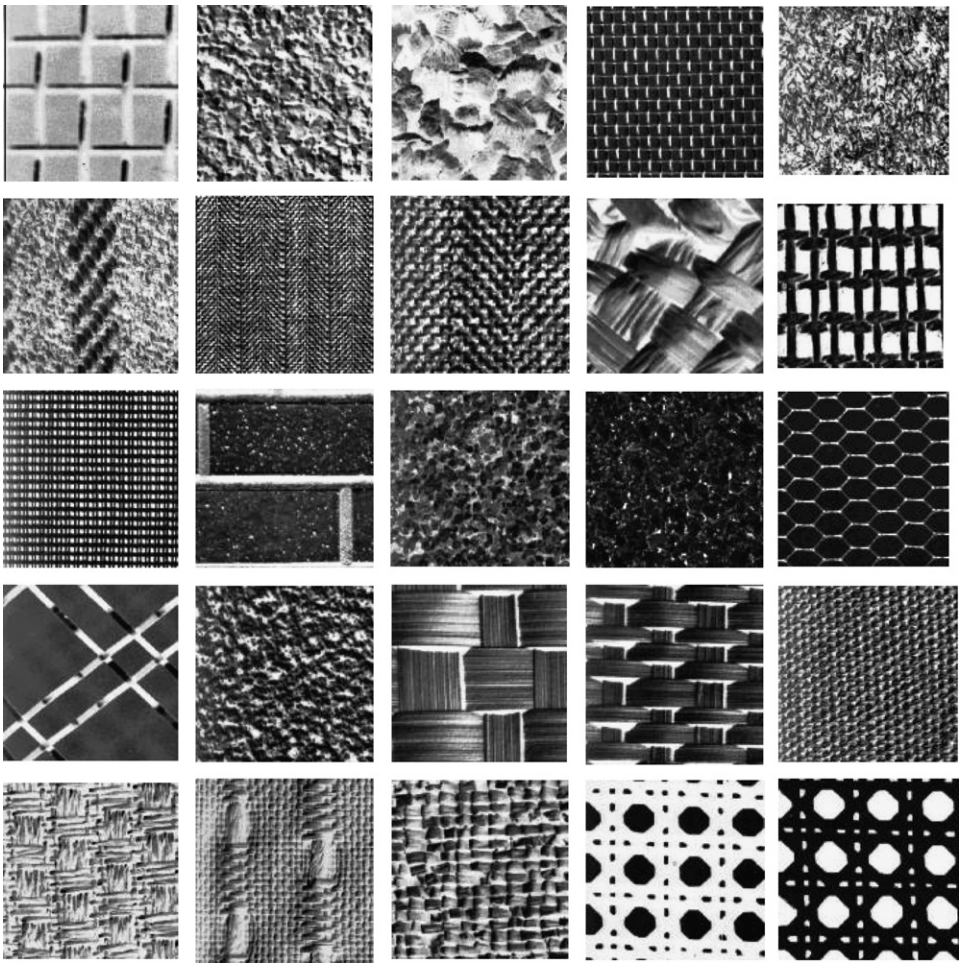


Fig. 1. Brodatz texture images. From left to right and top to bottom: D1, D4, D5, D6, D9, D11, D16, D17, D18, D20, D21, D26, D29, D32, D34, D47, D57, D64, D65, D77, D82, D83, D84, D101, D102.

broad variety of different realizations and implementations. The techniques applied to pattern recognition use artificial intelligence approaches (Avci, Turkoglu, & Poyraz, 2005c; Turkoglu, Arslan, & Ilkay, 2003).

2.2. Discrete wavelet transform

Recently, wavelet transforms are rapidly surfacing in fields as diverse as telecommunications, radar target recognition, texture image classification (Avci & Turkoglu, 2003). The main advantages of wavelets is that they have a varying window size, being wide for slow frequencies and narrow for the fast ones, thus leading to an optimal time–frequency resolution in all frequency ranges. Furthermore, owing to the fact that windows are adapted to the transients of each scale, wavelets lack of the requirement of stationary (Avci, Turkoglu, & Poyraz, 2005a, 2005c).

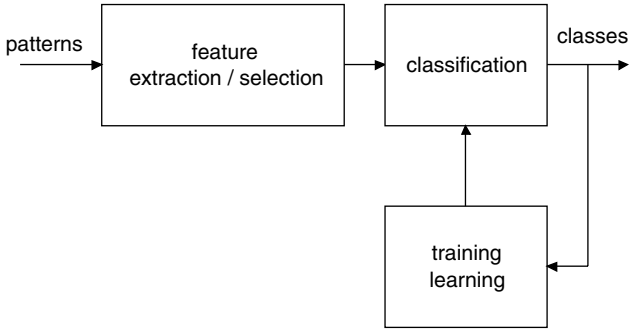


Fig. 2. The pattern recognition approach.

At using in image area of the wavelet, image is decomposed i.e., divided into four sub-bands and sub-sampled by applying DWT as shown in Fig. 3(a). These sub-bands are named as L-H1, H-L1 and H-H1 that represent the finest scale wavelet coefficients of detail images, sub-band L-L1 correspond to low frequency level coefficients of approximation image. The sub-band L-L1 alone is further decomposed to obtain the next coarse level of discrete wavelet coefficients (Arivazhagan & Ganesan, 2003).

For a two-level DWT decomposition as shown in Fig. 3(b). L-L2 will be used to obtain further decomposition. This decomposition process continues until final scale is reached. Coefficients obtained from DWT of approximation and detail images (sub-band images) are basic features that are shown here as useful for texture classification. Micro-textures and macro-textures are statistically

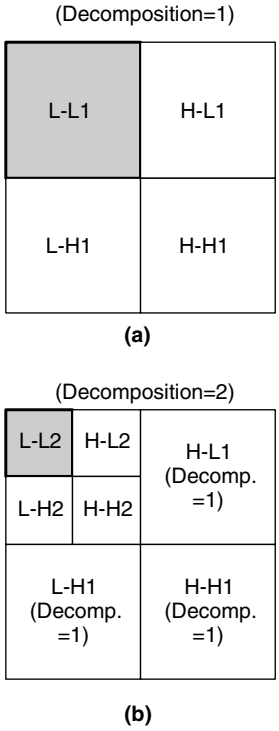


Fig. 3. Image decomposition, (a) one level, (b) two level.

characterized by using the features in approximation and detail of DWT. Namely, the values of the L-L, H-L, L-H, and H-H sub-band images or combinations of these sub-bands or the obtained features from these sub-bands very good characterize a texture image.

Readers may find information about wavelet families used in this study at MATLAB 5.3.

2.3. Wavelet Neural Networks

Neural Networks are systems that are constructed to make use of some organizational principles resembling those of the human brain (Avci, Turkoglu, & Poyraz, 2005b). They represent the promising new generation of information processing systems. Neural Networks are good at tasks such as pattern matching and classification, function approximation, optimization and data clustering, while traditional computers, because of their architecture, are inefficient at these tasks,especially pattern-matching tasks (Turkoglu et al., 2003). As for Wavelet Neural Networks try to combine aspects of the wavelet transformation for purpose of feature extraction and selection with the characteristic decision capabilities of neural network approaches (Avci et al., 2005b). The Wavelet Neural Network (WNN) is constructed based on the wavelet transform theory (Avci et al., 2005c) and is an alternative to feed-forward neural network (Avci et al., 2005c). Wavelet decomposition (Avci et al., 2005c) is a powerful tool for non-stationary signal analysis. Let x(t) be a piecewise continuous function. Wavelet decomposition allows one to decompose x(t) using a wavelet function $\Psi : R^n \rightarrow R$. Based on the wavelet decomposition, wavelet network structure is defined by

$$y(x) = \sum_{i=1}^{N} W_i \Psi[D_i(x - t_i)] + b \tag{1}$$

where D_i are dilation vectors specifying the diagonal dilation matrices D_i, t_i are translation vectors, and the additional parameter b is introduced to help deal with non-zero mean functions on finite domains. An algorithm of the back-propagation type has been derived for adjusting the parameters of the WNN (Avci et al., 2005c). Applications of Wavelet Neural Network in the medical field include classification of coronary artery diseases (Avci & Turkoglu, 2003; Avci et al., 2005a; Avci et al., 2005c; Turkoglu et al., 2003), characteristics of heart valve prostheses (Avci et al., 2005c), interpretation of the Doppler signals of the heart valve diseases (Avci et al., 2005c), classifying bio signals (Avci et al., 2005c), ECG segment classification (Avci et al., 2005c); however, to date Wavelet Neural Network based on Adaptive Norm Entropy analysis of texture image classification is a relatively new approach.

3. The feature extraction methods used in this study

The feature extraction method suggested in this study consists of 4 stages. These stages are summarized as below.

3.1. Stage-1

In this stage, known texture images are trained. These images are decomposed by using DWT. Then, norm entropy, mean and standard deviation of approximation and detail sub-bands of three level decomposed texture images (i.e., L-L*i*, L-H*i*, H-L*i* and H-H*i*; for $i = 1,2,3$) are calculated as features by using the equations given in Eqs. (2)–(4), respectively. Then, obtained these features stared in features library.

$$\text{norm entropy(ne)} = \frac{\sum_{i,j=1}^{N} |t(i,j)|^p}{N} \tag{2}$$

where p is the power and must be such that $1 \leqslant p < 2$.

$$\text{mean} = \frac{1}{N^2} \sum_{i,j=1}^{N} t(i,j) \tag{3}$$

$$\text{standard deviation(sd)} = \sqrt{\frac{1}{N^2} \sum_{i,j=1}^{N} [t(i,j) - \text{mean}]^2} \tag{4}$$

where $t(i,j)$ is the transformed value in (i,j) for any sub-band(one of L-L*i*, L-H*i*, H-L*i*, and H-H*i*) of size $N \times N$ (Arivazhagan & Ganesan, 2003).

For any texture image, these features given above (upto *i*-level L-L*i*, L-H*i*, H-L*i*, and H-H*i* sub-bands ($i = 1,2,3$)) are computed. Then these features are stored in the features library. This features library is further used in texture

image classification stage. These features are named as Wavelet Statistical Features (WSF). Texture image classification is realized that yielded good classification result when using a combination of the above WSFs. In addition to, finding the co-occurrence matrix features of original texture image, approximation and detail sub-bands (i.e., L-L1, L-H1, H-L1 and H-H1) of 1-level DWT decomposed images is proposed for improve the correct classification rate further. These features are named as Wavelet Co-occurrence Features (WCF) (Arivazhagan & Ganesan, 2003). The different co-occurrence features such as inverse difference moment, contrast, energy, norm entropy, local homogeneity, cluster shade, cluster prominence and maximum probability, as suggested in Arivazhagan and Ganesan (2003). The formulas of co-occurrence matrix features used in this study are given in Eqs. (5)–(12), respectively:

$$\text{inverse difference moment} = \sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\text{Co}(i,j)}{|i-j|^2}, \quad i \neq j \tag{5}$$

$$\text{contrast} = \sum_{i=1}^{N}\sum_{j=1}^{N}(i-j)^2\text{Co}(i,j) \tag{6}$$

$$\text{energy} = \sum_{i=1}^{N}\sum_{j=1}^{N}\text{Co}^2(i,j) \tag{7}$$

$$\text{norm entropy} = \frac{\sum_{i,j=1}^{N}|\text{Co}(i,j)|^p}{N}, \quad 1 \leqslant p < 2 \tag{8}$$

$$\text{local homogeneity} = \sum_{i,j=1}^{N}\frac{1}{1+(i-j)^2}\text{Co}(i,j) \tag{9}$$

$$\text{cluster shade} = \sum_{i,j=1}^{N}(i-Px+j-Py)^3\text{Co}(i,j) \tag{10}$$

$$\text{cluster prominence} = \sum_{i,j}^{N}(i-Px+j-Py)^4\text{Co}(i,j) \tag{11}$$

$$\text{maximum probability} = \text{Max}[\text{Co}(i,j)] \tag{12}$$

where

$$Px = \sum_{i,j=1}^{N} i\text{Co}(i,j) \quad \text{and} \quad Py = \sum_{i,j=1}^{N} j\text{Co}(i,j)$$

In this study, combination of this WSF and WCF mentioned above will used for feature extraction.

3.2. Stage-2

This stage is texture classification stage. Here, the features of known texture image calculated at Section 3.1 are used for intelligent classification. The training parameters and the structure of the MLP used in this study are as shown in Table 1. These were selected for the best performance, after several different experiments, such as the number of hidden layers, the size of the hidden layers, value of the moment constant and learning rate, and type of the activation functions. In Stage-1, adaptive norm entropy is used for both of WSF and WCF.

Table 1
MLP architecture and training parameters

The number of layers	3
The number of neuron on the layers	Input: 76, Hidden: 80, Output: 25
The initial weights and biases	The Nguyen–Widrow method
Activation functions	Log-sigmoid
Training parameters	
Learning rule	Back-propagation
Adaptive learning rate	Initial: 0.0001
	Increase: 1.05
	Decrease: 0.7
Momentum constant	0.98
Sum-squared error	0.00000001

During Neural Network based on Adaptive Norm Entropy (WNN-ANE) algorithm learning process, the p parameter (norm entropy parameter for both WSF and WCF) is updated by using 0.1 increasing steps together with weights to minimize the error. The resultant entropy data were normalized by dividing with max norm entropy value. The structure of WNN-ANE algorithm for texture image classification is shown in Fig. 4.

3.3. Stage-3

This stage is texture testing stage. Here, the unknown texture image is decomposed using DWT. Then, a similar set of WSFs and WCFs are extracted and compared with the corresponding feature values stored in the features library using WNN-ANE trained network parameters. In this study, 600 texture image that have random variety scale (256×256, 128×128, 64×64, and 32×32) for each of 25 texture image were used for testing of the correct texture image classification.

4. Experimental results and discussion

Twenty five monochrome texture images each of size 512×512, obtained from Brodatz (http://www.ux.his.no/~traden/broadatz.html, 2005.) image database shown in Fig. 1. is used in this study. The experiments are conducted with these texture images. In these experiment applications, variety feature vectors are used for comparative analysis. These feature vectors consist of different wavelet families. One another aim of this study is comparating each of these wavelet families with others about correct classification performance.

Feature database used in this study is a total of 8500 image regions of 25 texture images, constituted by dividing each 512×512 texture image into non-overlapping $4{,}256\times256$, $16{,}128\times128$, $64{,}64\times64$, and $25{,}632\times32$ image regions and by extracting 36 WSFs and 40 WCFs, averaged over these 340 image regions. Here, for each of these image

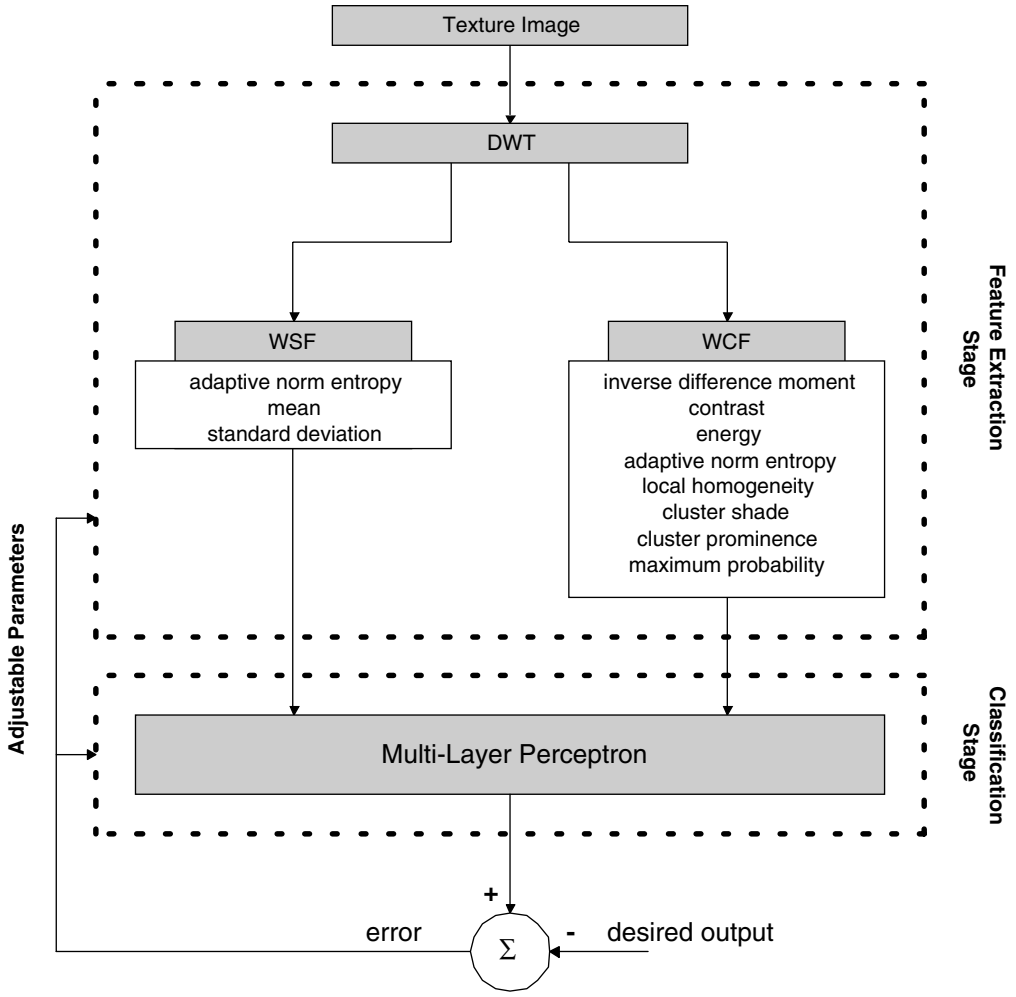


Fig. 4. The structure of WNN-ANE algorithm for texture image classification.

regions (i) 36 WSFs such as Adaptive Norm Entropy, mean and standard deviation of L-Li, L-Hi, H-Li and H-Hi (for $i = 1, 2, 3$) sub-bands of three-level DWT decomposed texture images and (ii) 40 wavelet co-occurrence features (WCF) such as inverse difference moment, energy, Adaptive Norm Entropy, local homogeneity, cluster shade, cluster prominence and maximum probability, derived from co-occurrence matrices, computed for variety angles (i.e., $\phi = 0°$, 45°, 90° and 135°) and averaged, of original texture images, approximation and detail sub-bands of 1-level DWT decomposed texture images. These 36 WSFs and 40 WCFs averaged over these 340 image regions for each of 25 image textures. These averaged features are used as feature database in these experimental applications.

Different feature vectors are obtained by using variety wavelet families for DWT decompositions of the texture images. These feature vectors can be given as below:

- *Feature vector and Classification-1 (FC-1)* = WSFs + WCFs + WNN-ANE (In this method, db 1 (daubechies 1) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-2 (FC-2)* = WSFs + WCFs + WNN-ANE (In this method, db 2 (daubechies 2) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-3 (FC-3)* = WSFs + WCFs+ WNN-ANE (In this method, db 3 (daubechies 3) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-4 (FC-4)* = WSFs + WCFs + WNN-ANE (In this method, bior 1.3 (biorthogonal 1.3) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-5 (FC-5)* = WSFs + WCFs + WNN-ANE (In this method, bior 2.2 (biorthogonal 2.2) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-6 (FC-6)* = WSFs + WCFs + WNN-ANE (In this method, bior 2.4 (biorthogonl 2.4) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-7 (FC-7)* = WSFs + WCFs + WNN-ANE (In this method, coif 1 (coiflets 1) wavelet filters were used for DWT decomposition)

Table 2
Results of texture classification using WSFs–WCFs–WNN-ANE (with 8500 image regions)

SL No.	Images	Correct classifications (%)											
		Feature vectors											
		FC-1	FC-2	FC-3	FC-4	FC-5	FC-6	FC-7	FC-8	FC-9	FC-10	FC-11	FC-12
1	D1	95.16	96.66	99.00	98.83	98.50	98.66	96.83	99.83	95.33	97.16	99.33	99.00
2	D4	94.50	97.00	96.83	98.33	97.83	98.16	96.16	99.50	98.66	99.83	98.16	99.16
3	D5	97.83	97.16	96.83	99.33	97.50	97.50	98.66	97.50	98.50	98.83	98.83	100
4	D6	98.83	100	98.33	96.00	98.83	99.00	100	94.16	99.83	97.00	99.83	98.16
5	D9	100	99.16	97.16	97.83	98.66	99.83	99.33	95.16	95.16	96.16	100	98.00
6	D11	98.33	99.66	96.66	95.00	98.00	100	97.66	99.33	100	99.33	97.00	99.50
7	D16	99.00	98.33	98.16	98.00	97.66	99.66	97.00	97.33	96.83	96.50	98.66	95.50
8	D17	99.66	98.83	98.50	98.50	98.16.	97.50	96.83	98.33	98.33	95.00	98.50	98.33
9	D18	97.66	97.50	99.83	99.83	98.16	97.00	99.00	96.83	97.66	99.00	99.83	99.00
10	D20	98.83	95.33	97.16	96.66	98.16	99.50	98.33	99.00	99.00	100	97.00	96.50
11	D21	99.16	100	98.00	96.16	99.50	100	100	97.83	98.83	98.83	98.16	98.16
12	D26	96.50	99.33	96.33	99.33	99.66	99.66	98.66	98.50	97.16	97.16	100	99.50
13	D29	96.00	100	94.00	98.33	97.83	98.00	99.66	97.66	99.33	96.50	98.33	98.50
14	D32	95.00	97.83	96.00	99.50	99.83	99.33	98.83	98.16	98.50	97.16	98.66	100
15	D34	95.33	98.50	95.66	99.33	98.83	98.83	98.00	98.50	97.16	98.66	100	99.16
16	D47	97.50	97.16	99.33	97.83	97.66	97.33	96.66	98.16	97.00	96.50	99.16	97.00
17	D57	98.00	99.66	99.50	96.16	99.16	97.83	95.50	95.50	98.16	99.83	98.16	99.33
18	D64	99.66	93.00	98.66	96.83	98.83	98.83	98.66	99.50	96.00	100	99.00	95.00
19	D65	92.83	96.00	97.00	96.50	96.66	98.50	98.33	97.83	96.00	99.50	99.83	96.33
20	D77	90.50	98.19	97.16	99.00	98.66	99.33	95.33	98.50	95.66	97.66	98.83	98.83
21	D82	94.66	98.50	97.16	98.66	96.66	96.66	99.50	100	99.50	97.50	96.83	98.00
22	D83	94.66	100	98.33	99.66	99.33	95.66	96.83	98.16	100	97.66	96.66	98.33
23	D84	100	97.33	99.83	97.16	99.00	98.50	99.33	96.50	98.16	99.16	97.66	98.83
24	D101	97.83	100	99.00	99.33	97.66	97.83	96.83	98.83	90.33	97.50	99.66	96.66
25	D102	97.83	90.83	99.16	98.16	100	96.00	99.16	98.50	98.33	98.16	99.16	98.66
Number of image regions correctly classified		14,552	14,673	14,662	14,702	14,765	14,752	14,707	14,709	14,684	14,692	14,804	14,733
Mean success rate		97.01	97.82	97.74	98.01	98.43	98.34	98.04	98.06	97.89	97.94	98.69	98.22

- *Feature vector and Classification-8 (FC-8)* = WSFs + WCFs + WNN-ANE (In this method, coif 2 (coiflets 2) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-9 (FC-9)* = WSFs + WCFs + WNN-ANE (In this method, coif 3 (coiflets 3) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-10 (FC-10)* = WSFs + WCFs + WNN-ANE (In this method, sym 2 (symlets 2) wavelet filters were used for DWT decomposition)
- *Feature vector and classification-11 (FC-11)* = WSFs + WCFs + WNN-ANE (In this method, sym 3 (symlets 3) wavelet filters were used for DWT decomposition)
- *Feature vector and Classification-12 (FC-12)* = WSFs + WCFs + WNN-ANE (In this method, sym 4 (symlets 4) wavelet filters were used for DWT decomposition)

From Table 2, it is found that when classification is carried out with WSFs + WCFs + WNN-ANE algorithm by using different feature vectors, which consist of variety wavelet families. Results of texture classification using wavelet statistical and co-occurrence features (with 8500 image regions) are given in Table 2. 600 texture image that have random variety scale (256×256, 128×128, 64×64, and 32×32) for each of 25 texture image were used for testing of the correct texture image classification.

5. Conclusion

In this study, it was developed an expert texture classification system for the interpretation of the texture images using pattern recognition methods. This WNN-ANE algorithm has capability of successfully classification of 25 texture images by using variety scales (256×256, 128×128, 64×64, and 32×32) of these texture images. There, using of MLP as a classifier increased the correct classification performance of system. The tasks of feature extraction and classification were performed using the WNN-ANE algorithm. The stated results show that the proposed method can make an effective interpretation. The performance of the expert system was given in Table 2.

The feature choice was motivated by a realization that WNN-ANE essentially is a representation of a texture image at a variety of scales and resolutions. In brief, the wavelet decomposition has been demonstrated to be an effective tool for extracting information from the texture images. The proposed feature extraction method is robust against to scale changing at texture images.

The most important aspect of the expert system is the ability of self-organization of the WNN-ANE without requirements of programming and the immediate response of a trained net during real-time applications. These

features make the expert system suitable for automatic classification in interpretation of the texture images. These results point out the ability of design of a new expert texture recognition assistance system.

The recognition performances of this study show the advantages of this system: it is rapid, easy to operate, and not expensive. This system offers advantage in radar texture image recognition and medical texture image classification. Besides the feasibility of a real-time implementation of the expert system, by increasing the variety and number of texture image additional information (i.e., quantification of the data length) can be provided for texture image recognition.

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